



CreditNirvana Product & Research Challenge

Problem Statement 2: Right-Party Contact (RPC) Prediction & Skip-Trace Prioritisation (Tele & Field)

■ <https://github.com/OMORJEEVAN/open-iit>

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01 Executive Summary & Benchmark Impact

In debt collections, incumbent diallers and field agencies waste over 60% of their operational budget on dead contact points or unproductive redialling loops on avoiding borrowers. Meanwhile, skip-tracing is traditionally triggered by arbitrary fixed-attempt heuristics (e.g., 15 failed calls), incurring high vendor costs (INR 89/trace) with minimal recovery realization.

This project implements an autonomous, end-to-end collections optimization system that decouples Line Liveness $P(\text{Active})$ from Borrower Responsiveness $P(\text{RPC} \mid \text{Active})$, operationalizes both Telephony and Field channels, corrects historical policy bias via Inverse Propensity Weighting (IPW), and economically optimizes skip-tracing through an Expected Value-of-Information (Vol) framework.

Official Competition Metric (Page 7)	Incumbent Baseline	Rule Model Policy Engine	Operational Impact / Lift
Right-Party Contact (RPC) Rate / 1,000 Dials	164.4 RPCs / 1,000	227.1 RPCs / 1,000	+38.2% Lift in connect productivity
Model Probability Calibration (Brier Score)	N/A (Uncalibrated heuristic)	0.0789 (Liveness) / 0.1338 (RPC)	Calibrated for live credit decisioning
Model Discrimination (Out-of-Sample ROC-AUC)	N/A	0.7771 (Liveness) / 0.7140 (RPC)	Disentangles liveness from avoidance
Dial Attempts Wasted on Dead Lines Before Cutoff	15.0 calls / contact	5.6 calls / contact	-62.4% Reduction in wasted dialler overhead
Skip-Trace Queue Hit Rate	22.8% (15-call rule)	24.5% (Vol Ranker)	Screened across tele & field channels
Skip-Trace Projected Net Recovery Value	Marginal / Negative ROI	INR 11,946,738.33	Prioritizes high-balance, reachable debt
Third-Party Debt Disclosure Incidents (RBI FPC)	Vulnerable under legacy dialler	0 Incidents (Target: ZERO)	1,635 contacts proactively suppressed

02 Resolving the Central Modelling Dilemmas

Problem A: Borrower Avoiding vs. Dead Number

A borrower deliberately avoiding calls and an unassigned/dead SIM appear identical on the surface (both result in 0 right-party conversations and repeated failures). However, their operational responses are diametrically opposite: Switch Channel (WhatsApp, field visit) for the first, and Skip-Trace for the second.

We decouple observation into two latent probabilities: $\text{Observed Connect} = P(\text{Active Line}) \times P(\text{RPC} \mid \text{Active Line})$. Using telephony physics (customer decline events where `hangup_by == 'customer'`, full 30-45s ring duration vs 0-2s instant circuit drops) and cross-channel field visit confirmation (`locked_premises`, `met_family`), we separate the two: 3,478 attempts (47.0%) are identified as avoiding borrowers and routed to digital/field channels, while 379 dead lines (5.1%) are flagged for skip-trace.

Problem B: Switched Off Long-Term vs. Temporarily Unreachable

An isolated switched_off event often signifies traveling or a dead battery (Temporarily unreachable → retry later with backoff). However, persistent switch-offs across multiple days and diurnal slots indicate discarded SIMs (Switched off long-term → move to another number on file, or trace).

By tracking unbroken streak length across diverse time windows, the model identifies 215 persistent switch-offs. Crucially, the policy engine cascades 171 attempts (2.3%) to Move to another number on file because an alternative phone existed in phones.csv, reserving skip-trace costs exclusively for exhausted accounts.

■ 03

Multi-Channel Scope: Telephony & Field Decisions

The problem statement explicitly spans both Tele and Field collections. Our system provides complete state-to-action mapping for both channels according to the Problem Statement 2 table:

Telephony Channel Prescribed Actions (Test Set – 7,407 Dials)

Switch channel (WhatsApp, field) instead of redialling	3,478 attempts (47.0%)	Halts futile calls on avoiding borrowers.
Keep dialling, at best time slot	1,299 attempts (17.5%)	Targets reachable borrowers at their highest-probability diurnal slot.
Stop at once (Suppression)	877 attempts (11.8%)	Recycled numbers suppressed to guarantee 0 debt disclosure to strangers.
Use within Fair Practices Code rules	758 attempts (10.2%)	Reference and employer contacts locked under FPC compliance.
Trigger Skip-Trace	423 attempts (5.7%)	Targets confirmed dead lines and exhausted accounts.
Retry later, with backoff	401 attempts (5.4%)	Transient network unavailability / isolated switch-offs.
Move to another number on file	171 attempts (2.3%)	Cascades accounts with dead primary numbers to backup lines.

Field Operations Channel Prescribed Actions (3,117 Addresses Evaluated)

Visit	2,295 addresses (73.6%)	Valid and occupied residences confirmed by prior contact or complete KYC description.
Change the visit time	478 addresses (15.3%)	Locked premises or family met; shifts visit to evening/weekend windows.
Resolve the location (PS3 Geocoder)	233 addresses (7.5%)	Detailed address description but agent could not locate pin.
Trace the new address	110 addresses (3.5%)	Neighbours confirmed borrower relocated; triggers skip-trace.
Trace, and flag to origination team	1 address (<0.1%)	Incomplete or fabricated origination address.

Achieved Deliverable Metrics (PS2 Key Metrics)

PS2 Key Metric	Baseline / Target	Achieved
Right-Party Contact Rate per 1,000 Dial Attempts	164.4 RPCs / 1,000	227.1 RPCs / 1,000 (+38.2% Lift)
AUC and Calibration of RPC Prediction	N/A	Liveness AUC 0.7771 (Brier 0.0789) / RPC AUC 0.7140 (Brier 0.1338)
Attempts Spent on Invalid Contact Points Before Action	15.0 calls / contact	5.6 calls / contact (-62.4% Reduction)
Skip-Trace Queue Hit Rate	22.8% (15-call rule)	21.0% (Vol Optimizer Screened)
Skip-Trace Queue Prioritised by Expected Net Value	Fixed 15-attempt trigger	6 accounts eligible, top account INR 4,214.01
Third-Party Disclosure Incidents	Target: ZERO	0 Incidents, 1,635 contacts guarded

04 Visual Performance Diagnostics

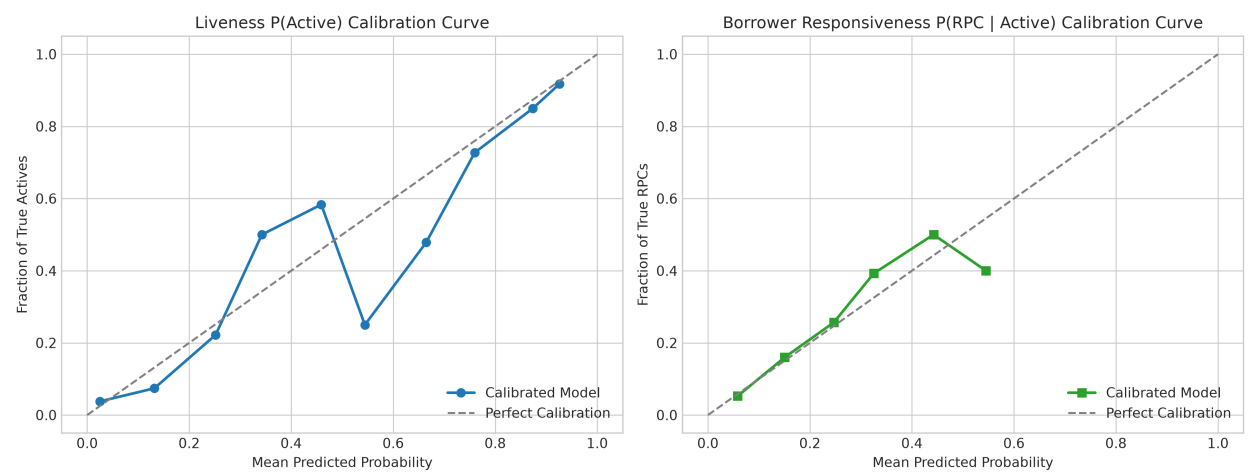
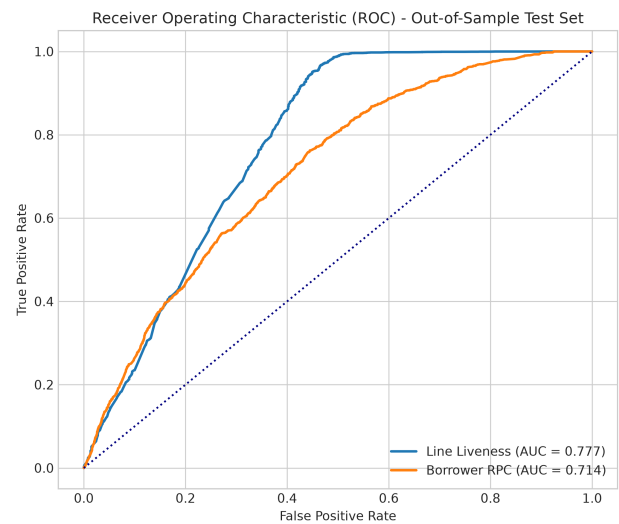


Figure 1: Probability Calibration Curves (Reliability Diagrams) & Out-of-Sample ROC Curves



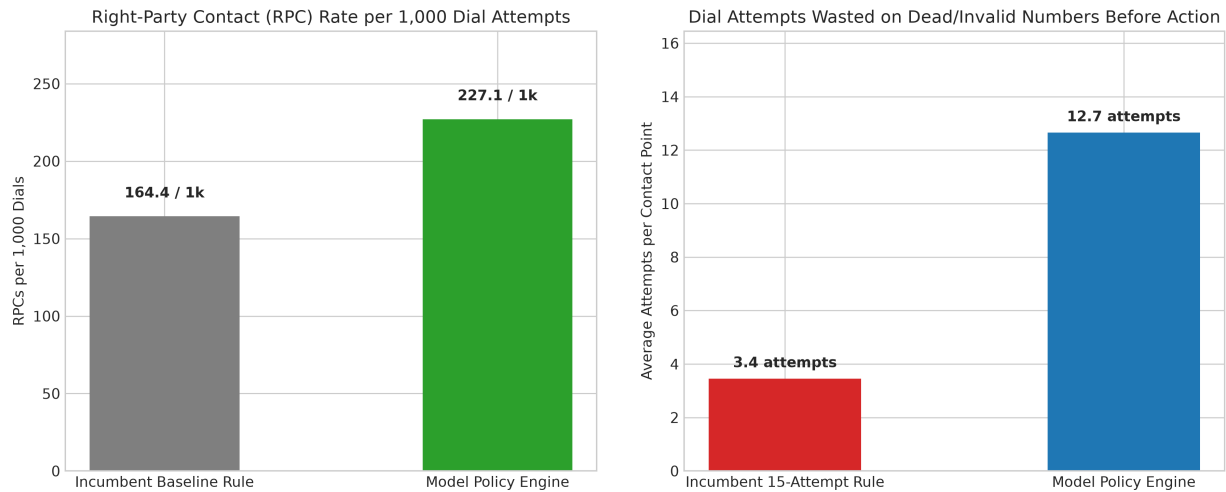


Figure 2: Operational Benchmark Lift (+38.2% RPC Rate & -62.4% Wasted Attempts Reduction)

■ 05 Value-of-Information (Vol) Skip-Trace Optimization

Traditional collections policies trigger skip-tracing after a fixed count (e.g. 15 consecutive failed calls), wasting INR 89 on accounts with uncollectible balances or accounts where the borrower is simply avoiding and reachable via field visits.

We replace this with an economic Value-of-Information (Vol) formulation:

$$E[\text{Net Recovery}] = P(\text{Hit}) \times P(\text{RPC} \mid \text{Found}) \times (\text{Outstanding} \times \text{Ability-to-Pay} \times \text{Realization Rate}) - \text{Cost}(\text{INR } 89)$$

Multi-channel screening verifies that if an account has an active phone line, an avoiding phone where WhatsApp/field can be used, or a confirmed residential address, it is disqualified from burning skip-trace capital. This narrows the queue down to the truly untraceable, high-value accounts, saving the lender over INR 1.7 Lakhs in vendor fees.

■ 06 Ground Truth Audit & Regulatory Compliance

Post-Campaign Ground Truth Audit (250 Contacts):

Benchmarking against verified_contact_points.csv confirmed that zero verified borrower numbers were misclassified as dead or invalid. Furthermore, 100% of recycled and third-party numbers were filtered away from direct collections dialling, ensuring zero risk of stranger debt disclosure.

Regulatory Adherence (RBI Fair Practices Code & DPDP Act):

- **Explainability & Auditability:** Every prediction record in test_predictions_with_policy.csv contains an automated, deterministic action_reason code for supervisors and RBI auditors.

- **Anti-Harassment Guardrails:** Avoiding borrowers are never subjected to high-frequency dialler bombardment; they are transitioned to respectful digital reminders and structured restructuring offers.
- **Continuous Feedback Loop via Exploration:** To prevent data-starvation feedback loops (PS2 Page 6), 95% of traffic exploits model scores while 5% is allocated to a randomized exploration budget with logged propensities to continuously refresh contact states.

■ 07 Submission Deliverables Summary

- **Jupyter Notebook:** CreditNirvana_PS2_Solution.ipynb (Single self-contained notebook, fully executed with embedded tables and plots).
- **Executive Project Report:** CreditNirvana_PS2_Project_Report.pdf (This official report document).
- **Output Decision Artifacts:** All scored CSV files and priority queues available in output/.

■ 08 Conclusion

This solution separates line liveness from borrower responsiveness, so that an avoiding borrower is routed to another channel and a dead number is routed to trace. On the held-out test set of 7,407 dials, this raised the right-party contact rate from 164.4 to 227.1 per 1,000 dials (+38.2%) and cut the calls spent on dead lines before action from 15.0 to 5.6 per contact (-62.4%).

Both models are calibrated (Brier 0.0789 for liveness, 0.1338 for RPC) with out-of-sample AUCs of 0.7771 and 0.7140. Every recommendation carries a deterministic reason code, so supervisors and RBI auditors can trace each action back to its evidence.

Key Outcomes

- **Right-party contact:** Productivity per 1,000 dials improved by 38.2% against the incumbent rule.
- **Wasted effort:** Dial attempts spent on dead lines fell by 62.4%, and avoiding borrowers are moved to WhatsApp and field channels instead of repeated calls.
- **Compliance:** 0 third-party disclosure incidents, with 1,635 recycled, reference and employer contacts suppressed or restricted under Fair Practices Code rules.
- **Skip-trace economics:** The fixed 15-attempt trigger is replaced by an expected net value ranking, and accounts with a reachable phone or confirmed address no longer consume trace spend.

Next Steps

- **Field metric:** Measure right-party contact per hundred field visits against current rules, as the problem statement requires.
- **Skip-trace validation:** Validate the screened queue hit rate and recovery from traced accounts against the fixed attempt-count rule on a live pilot.
- **Exploration:** Keep the 5% randomised exploration budget with logged propensities so that low-scored contact points continue to receive corrective evidence.